

Technical Paper for Prediction of Gait Kinematics with Gaussian Process Regression

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1 Introduction

In this supplementary material, we introduce the Gaussian Process Regression (GPR) algorithm for gait kinematics prediction in more detail and describe some of its practical aspect.

1.1 Gaussian Process Regression

A Gaussian Process (GP) is a collection of random variables, any finite number of which has a joint Gaussian distribution (Rasmussen, 2006). As one of stochastic processes, GPs are usually defined over time. However, when GPs are applied for regression, GPs are defined over the domain of the function to be estimated. Since it uses the Bayesian framework, GPR has two advantages; a principled way of model selection is provided and the probability density of predicted value is produced. For more details and tutorials on GP, refer the book of Rasmussen (2006) or the website (Rasmussen, 2011).

1.2 Contents of this material

First, we describe the GPR algorithm to predict *joint motions* from Section 2 through 4, and then explain the algorithm to predict *gait period* in section 5.

2 Definition of Training Set

For the prediction of normalized joint motions, training input vector $\mathbf{x}$, training output scalar y , and their sets $\mathbf{X}$ and $\mathbf{y}$ are defined as (1).

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_{N \times T}^\top \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1^\top & t_1 \\ \vdots & \vdots \\ \mathbf{b}_1^\top & t_T \\ \mathbf{b}_2^\top & t_1 \\ \vdots & \vdots \\ \mathbf{b}_N^\top & t_T \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_{N \times T} \end{bmatrix} \quad (1)$$

where $\mathbf{b}$ is a vector denoting subject's body parameter and N is the total number of subjects, t denotes a time index in the normalized time frame, T is the last index of the normalized time frame, y is a gait pattern value (given as a distance or an angle) corresponding to its $\mathbf{x}$. For example, if a training set is made for ankle flexion pattern, $\mathbf{x}_{(i-1) \times T + j}$ includes the body parameter $\mathbf{b}_i$ of i -th subject and the j -th time index t_j , and corresponding $\mathbf{y}$ is ankle flexion angle for the given condition. This training set is built for all of the 14 joint motions that represent gait kinematics.

Practically, all of training data set are shifted and re-scaled before GPR to have standard normal distribution whose mean is zero and standard deviation is 1. It helps to make a mean function of GP and to obtain a more accurate optimal solution with a numerical optimization program.

3 Design of Gaussian Process model

A GP is completely specified by its mean function and covariance function (Rasmussen, 2006), as shown in (2):

$$f(\mathbf{x}) \equiv \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (2)$$

where mean function $m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$, and covariance function $k(\mathbf{x}, \mathbf{x}') = \mathbb{E}[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))]$.

In our GP model, the mean function and the covariance function are determined as (3) and (4)

$$m(\mathbf{x}) = \nu_m \quad (3)$$

$$k(\mathbf{x}, \mathbf{x}') = \nu_0 \exp \left\{ -\frac{(\Delta \mathbf{b})^\top \Lambda_b (\Delta \mathbf{b}) + \lambda_t \Delta t^2}{2} \right\} + \nu_1 \delta(i, j) \quad (4)$$

where $\Delta \mathbf{b}$ is defined as $\mathbf{b} - \mathbf{b}'$ and Δt is defined as $t - t'$. $\delta(i, j)$ is Kronecker delta function. Λ_b is a diagonal matrix whose elements are $\lambda_{b1} \dots \lambda_{b14}$. Hyperparameter set is defined as a set of $\nu_m, \nu_0, \nu_1, \lambda_b, \lambda_t$, and denoted as Θ . This hyperparameter set determines the characteristics of GP. Practically, without loss of generality, we can set the mean function in (3) as zero-constant because we shifted $\mathbf{x}$ to have zero mean in Section 2.

Comparing with the general covariance function of GPs such as squared exponential function, one remarkable point is that we assume Separable model in GP to separate the effect of two independent variables, or body parameters $\mathbf{b}$ and time t as shown in (4). This Separable model is widely used for the analysis of multivariate spatial data and spatiotemporal data, like our problem.

Then, based on the assumption of the Gaussian process prior, the joint density for the observations $\mathbf{y}$ and a target value y^* to be predicted for a new input vector $\mathbf{x}^*$ is expressed as:

$$\begin{bmatrix} \mathbf{y} \\ y^* \end{bmatrix} \sim \mathcal{N} \left(\mathbf{0}_{NT+1}, \begin{bmatrix} \mathbf{K} & \mathbf{k}_* \\ \mathbf{k}_*^T & \kappa \end{bmatrix} \right) \quad (5)$$

where $\mathbf{0}_{NT+1}$ is a vector of the size $NT+1$ whose elements are all 0's, $\mathbf{K}$ is a matrix whose elements K_{ij} is a covariance function value of $k(\mathbf{x}_i, \mathbf{x}_j)$. $\mathbf{k}_* = [k(\mathbf{x}^*, \mathbf{x}_1) \dots k(\mathbf{x}^*, \mathbf{x}_{NT})]^T$. and $\kappa = k(\mathbf{x}^*, \mathbf{x}^*)$

By conditioning the joint density on the observations $\mathbf{y}$ (referring to Appendix A.2 in Rasmussen (2006)), we can obtain the Gaussian Process posterior (the predicted probability distribution) of joint motion y^* for a given test vector $\mathbf{x}^*$ including body parameter $\mathbf{b}^*$ and time t^* as (6).

$$p(y^* | \mathbf{x}^*, \mathbf{X}, \mathbf{y}, \Theta) = \mathcal{N}(\mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{y}, \kappa - \mathbf{k}_*^T \mathbf{K}^{-1} \mathbf{k}_*). \quad (6)$$

4 GP Model Optimization

The newly devised GP model for gait pattern prediction is defined by mean function (3) and covariance function (4), and their characteristics are adjusted by the hyperparameter set Θ . ν_0, ν_1, λ_b and λ_t are the scale factors for training input vector $\mathbf{x}$, body parameters $\mathbf{b}$, time index t and measurement noise respectively.

By selecting the hyperparameter set, the optimal GP model which most well represents all of training set is selected. The optimization of hyperparameter is performed by maximizing likelihood among training dataset. Practically, the optimization is performed by maximizing log-likelihood as

shown in (7).

$$\log p(\mathbf{y}|\mathbf{X}, \Theta) = -\frac{1}{2}\mathbf{y}^\top \mathbf{K}^{-1}\mathbf{y} - \frac{1}{2}\log |\mathbf{K}| - \frac{NT}{2}\log 2\pi. \quad (7)$$

In this study, the optimization was performed by a line search algorithm (Press et al., 2009). The partial derivate of $\log p(\mathbf{y}|\mathbf{X}, \Theta)$ with respect to a hyperparameter Θ , which is required to use a gradient-based optimization method, is obtained trivially as

$$\frac{\partial \log p(\mathbf{y}|\mathbf{X}, \Theta)}{\partial \theta} = \frac{1}{2}\mathbf{y}^\top \mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta} \mathbf{K}^{-1}\mathbf{y} - \frac{1}{2}\text{tr}(\mathbf{K}^{-1} \frac{\partial \mathbf{K}}{\partial \theta}). \quad (8)$$

5 Gait period prediction

Comparing with prediction of joint motion, prediction of gait period is relatively simple because its training set does not include time index. For prediction of gait period, the training data and the covariance function are set as (9) and (10).

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1^\top \\ \vdots \\ \mathbf{b}_N^\top \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \quad (9)$$

$$k(\mathbf{x}, \mathbf{x}') = v_0 \exp \left\{ -\frac{(\Delta \mathbf{b})^\top \Lambda_b (\Delta \mathbf{b})}{2} \right\} + v_1 \delta(i, j) \quad (10)$$

Here, y_i is i -th subject's average period. All the other GPR procedures are performed in the same way as described in the previous part. Note that the number of observations in the period prediction is N rather than NT .

In the same way as in the section 3, a probability distribution of a period y^* for a given test set $\mathbf{x}^*$ is obtained as:

$$p(y^*|\mathbf{x}^*, \mathbf{X}, \mathbf{y}, \Theta) = \mathcal{N}(\mathbf{k}_*^\top \mathbf{K}^{-1}\mathbf{y}, \kappa - \mathbf{k}_*^\top \mathbf{K}^{-1}\mathbf{k}_*), \quad (11)$$

where $\mathbf{K}$ is a matrix whose elements K_{ij} is a covariance function value of $k(\mathbf{x}_i, \mathbf{x}_j)$. $\mathbf{k}_* = [k(\mathbf{x}^*, \mathbf{x}_1) \dots k(\mathbf{x}^*, \mathbf{x}_N)]^T$. and $\kappa = k(\mathbf{x}^*, \mathbf{x}^*)$. Likewise, the log likelihood for model selection is obtained as:

$$\log p(\mathbf{y}|\mathbf{X}, \Theta) = -\frac{1}{2}\mathbf{y}^\top \mathbf{K}^{-1}\mathbf{y} - \frac{1}{2}\log |\mathbf{K}| - \frac{N}{2}\log 2\pi. \quad (12)$$

6 Prediction Result

This section shows two prediction results based on the theories which explained in the previous sections. Regarding the database and the subject's body parameters, refer to the manuscript.

References

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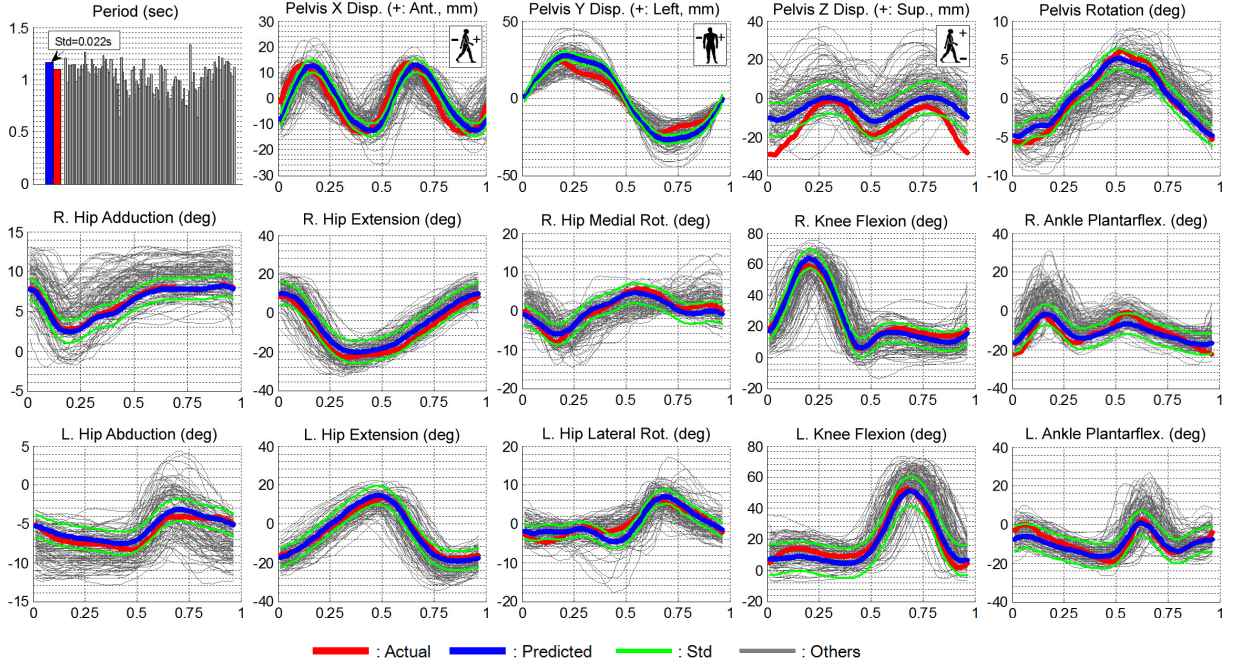


Figure 1: Gait pattern prediction result using GPR with subject S1's body parameters

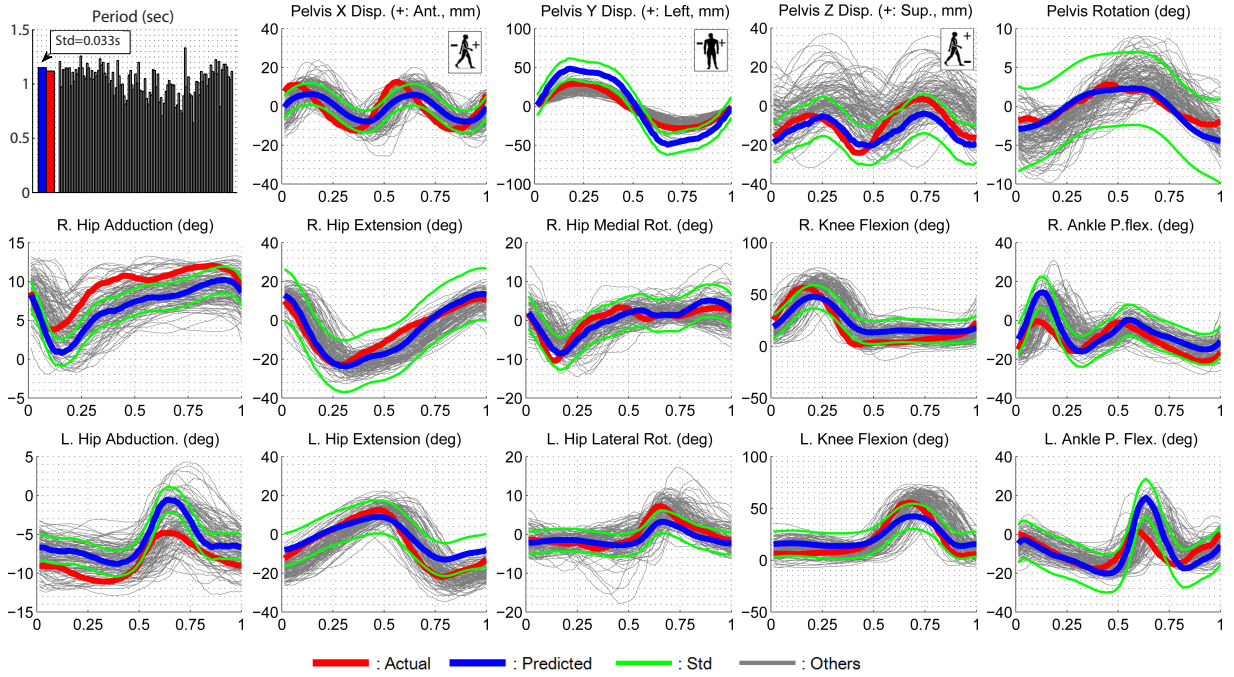


Figure 2: Gait pattern prediction result using GPR with subject S2's body parameters